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# =====
# Titanic Dataset EDA
# =====

# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("Titanic-Dataset.csv")

# -----
# 1. Display first rows
# -----
print("First 5 Rows:")
print(df.head())

# -----
# 2. Dataset Information
# -----
print("\nDataset Information:")
print(df.info())

# -----
# 3. Missing Values
# -----
print("\nMissing Values:")
print(df.isnull().sum())

# -----
# 4. Summary Statistics
# -----
print("\nSummary Statistics:")
print(df.describe())

# -----
# 5. Median
# -----
print("\nMedian:")
print(df.median(numeric_only=True))

# -----
# 6. Standard Deviation
# -----
print("\nStandard Deviation:")
print(df.std(numeric_only=True))

# -----
# 7. Histograms
# -----
df.hist(figsize=(12,10))
plt.suptitle("Histogram of Numerical Features")
plt.show()

# -----
# 8. Boxplots
# -----
numeric_columns = df.select_dtypes(include=['int64', 'float64']).columns

plt.figure(figsize=(15,8))
for i, col in enumerate(numeric_columns):
    plt.subplot((len(numeric_columns)+2)//3,3,i+1)
    sns.boxplot(y=df[col])
    plt.title(col)

plt.tight_layout()
plt.show()

# -----
# 9. Correlation Matrix
# -----
plt.figure(figsize=(10,8))
corr = df[numeric_columns].corr()

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sns.heatmap(corr,
             annot=True,
             cmap='coolwarm',
             fmt=".2f")

plt.title("Correlation Matrix")
plt.show()

# -----
# 10. Pairplot
# -----
sns.pairplot(df[numeric_columns])
plt.show()

# -----
# 11. Survival Count
# -----
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()

# -----
# 12. Survival by Gender
# -----
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()

# -----
# 13. Survival by Passenger Class
# -----
plt.figure(figsize=(6,4))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title("Survival by Passenger Class")
plt.show()

# -----
# 14. Age Distribution
# -----
plt.figure(figsize=(8,5))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()

# -----
# 15. Fare Distribution
# -----
plt.figure(figsize=(8,5))
sns.histplot(df['Fare'], bins=30, kde=True)
plt.title("Fare Distribution")
plt.show()

print("\nEDA Completed Successfully!")
```

